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The Water Supply of the Desert Fortresses in the Jordan Valley

By *Günter Garbrecht and Yehuda Peleg*

One of the prerequisites for the existence of towns, settlements, and even military installations such as fortresses is the guarantee of a secure water supply. It is essential for the survival of the inhabitants. The water for daily consumption must be supplied mostly from the surrounding geographical area. Difficulties arise most frequently from the fact that supply and demand are regulated by completely different factors and so only rarely coincide. The demand for water is influenced by the prevailing customs of the inhabitants, while available water supply is ruled by climate, precipitation, and the topographic conditions of the site.

The conditions at fortresses in arid areas are extremely unfavorable. Precipitation is minimal and very unevenly distributed throughout the year; and fortresses are situated on isolated hills or rocky peaks for strategic purposes. Securing the water supply under these circumstances calls for much ingenuity and technical expertise on the part of the engineer. Nevertheless, the second-first century BCE Hasmonean builders were able to solve these problems in the desert fortresses around the Jordan Valley.

Local Conditions

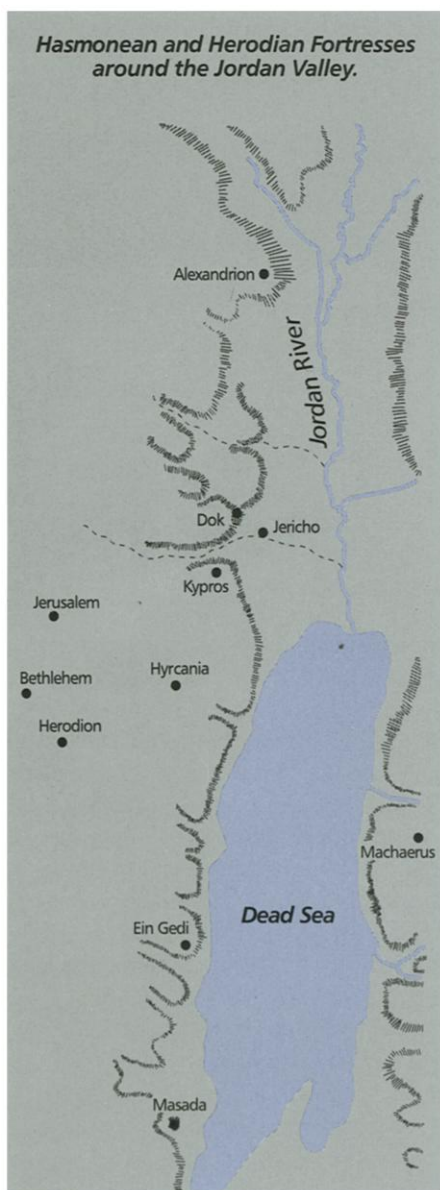
The watershed between the Mediterranean and the Jordan Valley runs in an

approximately north-south direction at approximately 800 to 900 m above sea level, through Jerusalem, Bethlehem, and Hebron. The Hasmoneans and Herod the Great built a number of fortresses in the rugged mountain area which descends steeply to the east of this watershed, in the rain shadow of the mountains. This arid area east of Jerusalem, called the Judaeen Desert, was only sparsely populated by nomadic herders

Fortress Dok on Gebel Karantal, the only Hasmonean fortress not built up by Herod. (The wall on top of the mountain is modern). *All photographs and illustrations courtesy of the authors.*

throughout antiquity. This desert, however, served as the perennial retreat of political refugees. They waited there for more propitious circumstances, gathered their supporters, and sometimes returned in victory to Jerusalem. The most famous of these refugees was David:

And David went up from thence, and dwelt in strongholds at Engedi. And it came to pass, when Saul was returned from following the Philistines, that it was told him, saying, Behold, David is in the wilderness of Engedi. Then Saul took three thousand chosen men out of all Israel, and went to seek David and his men upon the rocks of the wild goats. (I Sam 23:29–24:2)



Refugees such as David, who attained power after they had come to know the desert from personal experience, were probably the most inclined to found well-designed refuge fortresses in the desert. In the beginning, these were simple strongholds. Some were reinforced and occasionally equipped with many conveniences and the trappings of royalty (Plöger 1955; Harder 1962).

The period from the Seleucid conquest of Palestine in 201 BCE to the Jewish War of 66–70 CE was a politically volatile period of change and revolution. The Maccabean revolt of 167 BCE brought about the independence of Judea, which lasted until the Roman occupation of the country in 63 BCE.

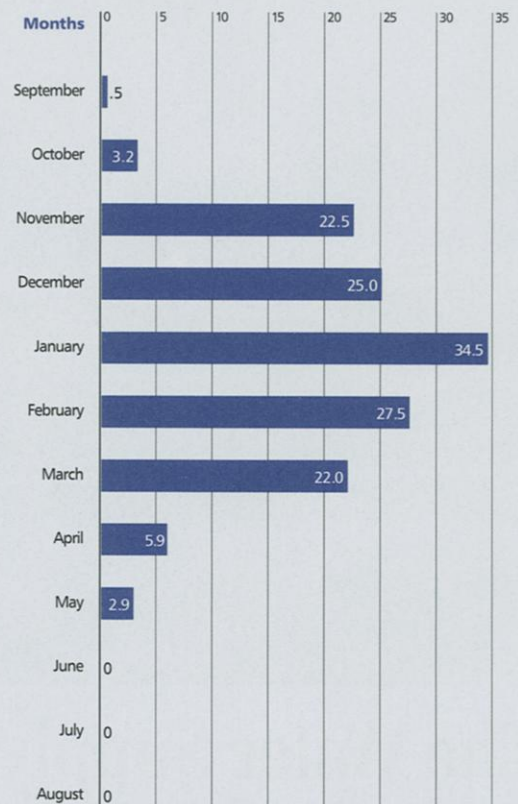
Jewish and Roman procurators, sometimes with the title of King or Queen, governed a region ravaged by political and religious tensions. During such times the desert fortresses proved to be essential strongholds and places of refuge for rulers in difficulty. Since none could be sure for any length of time of the strength of his or her position, these fortresses were valuable retreats in which to wait for the suppression of an uprising or the arrival of help from outside. For instance, during the Parthian invasion, King Herod left his family at Masada while he went to ask the Romans for assistance (Josephus *Jewish War* 1.13. 8).

The main desert fortresses were Alexandrion, Dok, Cypros, Hyrcania, Herodion, Masada, and Machaerus. All these installations were erected in excellent

Precipitation in Jericho, showing concentration in winter months.

Distribution of precipitation is actually much more erratic than the averages shown below.

Average precipitation in millimeters



defensive positions on mountain tops or the nearly inaccessible spurs of mountain ranges. A stronghold, however, must have not only an impregnable position, but also the capability to withstand a long siege. The clear military advantage of the position on a mountain top in the desert represents a serious handicap with respect to water supply.

The mean annual precipitation on the western slopes of the Jordan Valley is 50 to 200 mm, with the amount increasing from south to north. The precipitation falls mostly as rain, almost entirely between October and April, most of it in December, January, and February. The statistics of mean annual precipitation, however, can be misleading, since there are years in which only a few millimeters of rain falls, and such years can occur consecutively. Converse-

ly, in other years the precipitation can reach double the average. In most instances, this rainfall comes in short, heavy showers. Since the hard and barren desert soil absorbs only a little water, most of the precipitation runs off in sudden wadi floods, which sometimes assume catastrophic proportions. Despite this situation, the geology of the Judaeen mountains and the rainfall upon the western slopes foster the emergence of a number of springs on their eastern slope, for instance Ein Gedi, or Elisha's spring at Jericho.

Water Resources

The fortresses and military strongholds in the Judaeen mountains could draw on the following resources for their water supply: sporadic runoff, springs, and ground water. Runoff from the infrequent rain showers could either be diverted from natural collecting channels (the wadis) or—under suitable topographic conditions—collected in artificial channels. In view of the infrequent occurrence and very unpredictable amount of

the precipitation, storage was an essential element of any water-supply system. Water collected in the winter had to be stored for use during at least 12 and up to 24 months. Under such circumstances, only large underground cisterns were suitable for the desert fortresses. There are a significant number of springs on the eastern slope of the Judaeen mountains, fed by the more significant rainfall west of the watershed (e.g., 520 mm in Jerusalem). The output of the springs varies with the season, but the flow rarely ceases altogether. The variations range from $Q_{min} : Q_{max} = 1:1.4$ (Q = volume rate of flow) at Ein Elisha near Jericho to $Q_{min} : Q_{max} = 1:16$ at Ein Auja, north of Jericho. In principle, it is possible to tap the west-to-east descending aquifer with deep enough wells, and so to exploit this ground water directly.

Water Demand

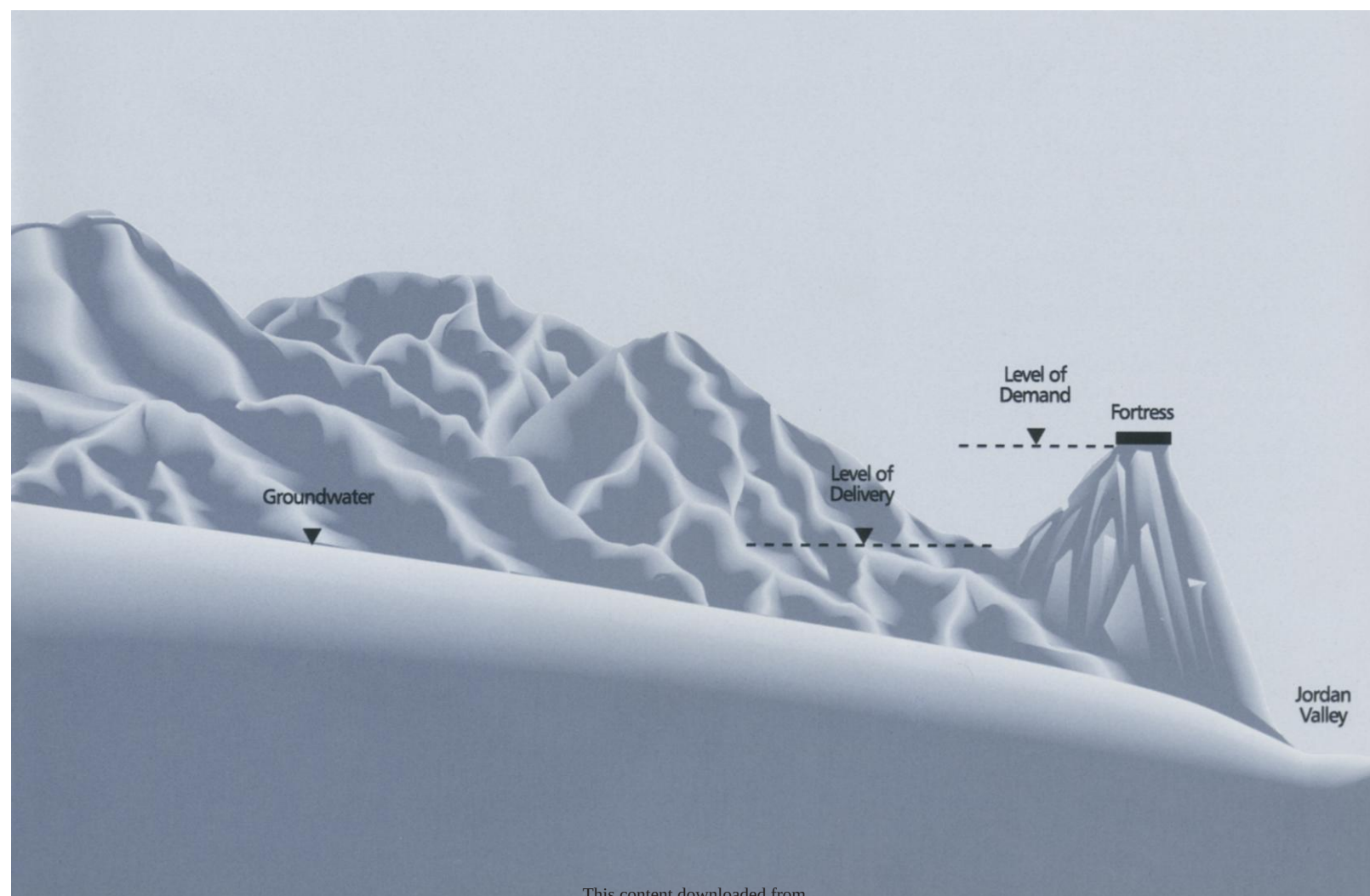
Water was needed in the fortresses mainly for drinking and washing for the permanent administrative, supply, and

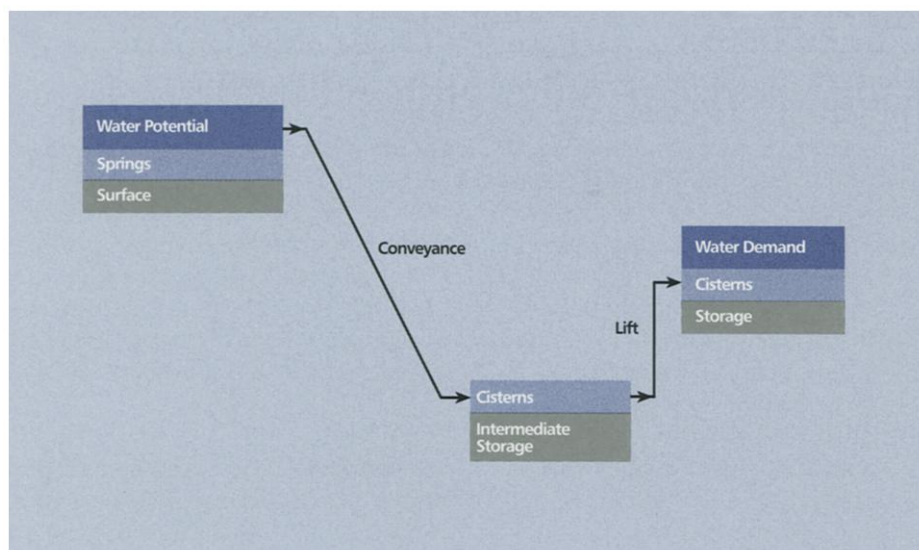
maintenance staff, as well as for the military units stationed there at any one time. It is impossible to estimate how much water was needed in any fortress, as the number of people and animals to be supplied is not known. A minimum of 10–20 liters a day for every inhabitant can be assumed under siege conditions. At Herod's fortress-palaces, daily water-use per person may have reached 200–300 liters and more. This water use was spread quite evenly throughout the year.

Water Use

As none of the fortresses occupied a large area, sufficient precipitation could not be collected inside the walls to fill cisterns large enough to assure the water supply for an extended period. In addition, the fortresses were built in the mountains, mostly on high peaks, and the groundwater table lay several hun-

Position of the fortresses (general sketch, not to scale), illustrating the topographical difficulty to supply the fortresses with water.





General scheme of the water supply systems.

dreds meters below them. This water could not be recovered with the technology available in the second or first century BCE. As a result, additional water had to be brought to such sites from outside. Since no waterlifting devices were available in antiquity which could lift significant amounts of water to very great heights, the water supply had to be conveyed in gravity-flow conduits. Because the fortresses had to be situated on high places for defensive reasons, only springs or surface runoff above the level of the supply channel could be used, and ground-level conduits could bring water only to the foot of the fortress hill. The level of the supply channel could be elevated by means of a bridge to the fortress hill or a ground-level pressurized conduit. As a result, the cisterns which were essential in equalizing the very widely fluctuating water supply had to be situated outside the fortress on the slopes of the hill. In peacetime, water was conveyed from this storage level to the consumption level in jars or water skins by porters or pack animals. During a siege, however, such cisterns could be reached only with difficulty, and sometime, not at all. Thus sufficient water-storage capacity had to be prepared inside the walls to allow for such an eventuality. These cisterns within the walls, which always had to be filled up from the lower, out-

side cisterns, were the mainstay of water supply during a siege.

Owing to the prevailing hydrological and topographical conditions, the following procedures were used in organizing the water supply of these fortresses :

- (1) Water catchment at springs, wadis, or collecting channels.
- (2) Conveyance of the water by a channel to the foot of the hill or, when a bridge or pressure con-



Channel leading to the cisterns on the eastern slope of Gebel Karantal.

duit was installed, to a point somewhere on its slope.

- (3) Intermediate storage on the slope of the fortress hill to store a supply sufficient to allow for different availability over time.
- (4) Transportation by pack animals into cisterns in the fort for storage and consumption. (For further details see Garbrecht/Peleg 1989 and Amit et al. 1989)

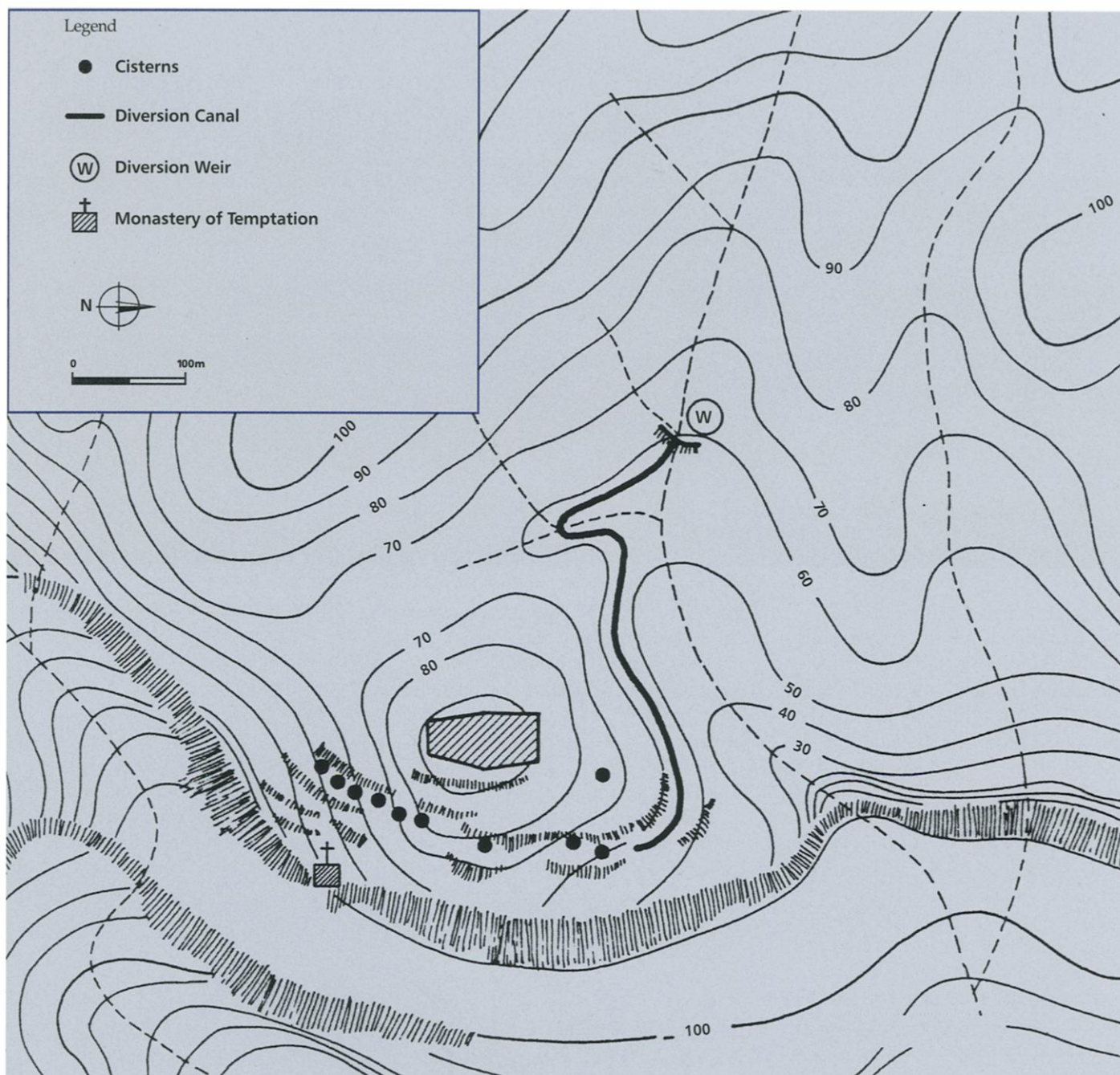
The fortress Dok provides a good example of a system supplied through diversion of water from a wadi. The system at Cypros will be described in some detail as an example for water collection and spring capture.

Dok

The fortress Dok (Dagon), on the Quarantana peak (Gebel Karantal) west of Jericho, is scarcely mentioned in historical sources. It was probably founded during the Maccabean uprising by Ptolemy, the son of Abubus, *strategos* at Jericho. Simon, the brother of Judas Maccabaeus, High Priest since 143 BCE was murdered there in 134 BCE. Dok is apparently the only fort that was not later rebuilt by Herod.

The east side of the Quarantana mountain is a sheer cliff descending to the Jericho plain, and in the north and south there are steep slopes. Only on the west is the fortress hill connected to the Judaeen mountain plateau, and here a dry moat was excavated. Not much is left of the fortress, as the peak was cleared at the beginning of this century for the building of a church, which was never completed.

Dok has the simplest water-supply system of the group of fortresses discussed here. During the infrequent rains, water was diverted from a wadi on the west to nine cisterns. One other cistern was situated somewhat higher. It seems that, owing to the local topography, even during a siege the cisterns on the steep slopes were accessible only to the defenders, and not to the assailants. The supply channel for this system began at a small diverting weir above a small cliff in the wadi. Only traces of masonry on the cliff show the position of the weir.



The first cistern is 500 meters away. The channel, partly built of field stones and partly hewn in the rock, is 0.50 to 0.70 m broad, with a fall of about 1%. When the channel was free of obstructions (gravel from the wadi or erosion material from the slopes) and flowing at a depth of 0.30 m, it could deliver up to 1.2 cubic meters per second. The cisterns filled by this channel are roughly hewn in the hard limestone bedrock. They are approximately rectangular in plan (L[ength] 7–11 m, W[idth] 3–5 m, H[eight] 5–7 m). Their ca-

pacities range from 190–270 cubic meters (m^3); the total capacity of the nine cisterns is 2100 (m^3). The lower part of the cistern openings, which were made quite high to facilitate the removal of the construction spoil, as later blocked by masonry to enlarge the cistern capacity. All the cisterns were lined with several layers of plaster.

Cypros

Cypros is situated 3 km south of Dok, on the prominent mountain peak Tel el

Gebel Karantal and fortress Dok, diversion weir, channel and cisterns according to Amit (1989). The occasional flow in the wadi was diverted into the channel and conveyed by it to the storage cisterns.

Akaba, 250 m above the Jericho plain. The fortress consists of a 1800 m^2 acropolis and a lower terrace of about 4500 m^2 to the south. The ruins are from the Hasmonean and Herodian periods. In the *Jewish War* (1. 21. 9), Josephus Flavius writes that Herod built an especially



strong and ornate fortress above Jericho, naming it Cypros in honor of his mother (Harder 1962). There is no literary evidence of a Hasmonean fort here, but it is assumed that one of the forts Trax and Taurus (mentioned by Strabo in his account of Pompeius' drive from Jericho to Jerusalem) occupied this site.

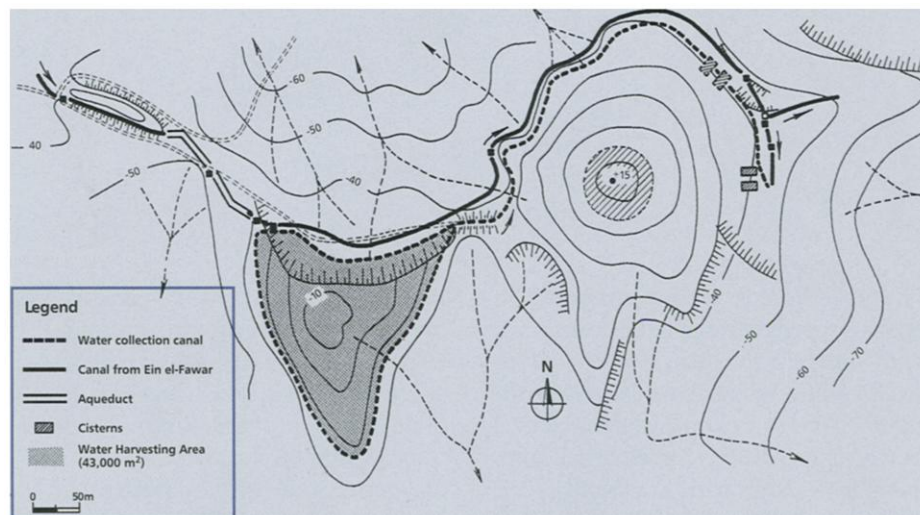
While the Hasmonean stronghold was a fortress proper, the Herodian installation was more of a palace, its luxury and comfort attested by the two sumptuous baths found there. The older Hasmonean fortress was supplied by a collecting channel. Three hundred meters southwest of Tel el Akaba there is a hill only 25 m lower than the acropolis. A channel at the height of the col (saddle) between the two peaks, 55 m below their summits, encircles the lower hill, an area of about 43,000 m², for collection of its run-off water. The channel most probably went around the north side of Tel el Akaba to the cisterns on the eastern slope. Together with the northern half of the fortress hill, the entire catchment area was about 100,000 m². The mean yearly precipitation near Jericho is 140 mm. If half of this total results in run-off and 50% of the run-off reaches the channel, approximately 3,500 m³ can be "harvested" yearly. The total capacity of the four cisterns on the eastern slope is

more than 2000 m³. The probable average annual supply from this source would have been more than enough to fill up the cisterns in the fort during the winter months. The cisterns are built at approximately 40 m below sea level, 5 m below the saddle, and approximately 55 m below the acropolis.

Apart from Herodion, which was supplied by spring water from the so-called "Solomonic Pools," Cypros was the

Cypros, seen from Jericho. Herod built this very strong and ornate fortress on a former Hasmonean fort and named it in honor of his mother. Cisterns on the slope are visible.

insufficient. As there were no springs, suitable wadis, or catchment areas in the immediate vicinity, the outflow of two springs in the upper part of the nearby Wadi Kelt, 7.5 and 12 km distant



only fortress to make use of spring water. It seems that after the rebuilding of the fort by Herod and its conversion into a luxurious castle, the former reliable but frugal water supply was considered to be

Water supply system of Cypros according to Meshel-Amit (1979). The collecting system served the Hasmonean fort. Herod built a most elaborate aqueduct in order to bring spring water to Cypros.

as the crow flies, was captured and led to Cypros through extremely difficult terrain. The output of the upper spring (Ein Fara) was led by a 6 km long channel to the lower spring (Ein el Fawar), and their combined output brought to Cypros. The channel from Ein el Fawar to Cypros was 14 km long and incorporated ten bridges and five tunnels in order to overcome the many small ravines descending into the wadi. The lower part of some of the pillars of the longest bridge (L 135 m), are still standing. The maximum height of the bridge was about 14 m, and the span of the widest arch was 12.60 m.

Since both Ein Fara and Ein el Fawar are karstic springs, their output varies greatly over time. At Ein el Fawar a maximal output of 0.83 cubic meters per second (m^3/s) was recorded, but sometimes the flow ceases completely. At a median output of 0.133 m^3/s the overall yearly total would be 4 million m^3 . Taking into account the considerable losses on the way, a yearly supply of 3 million m^3 can be expected at the end of the conduit, 0.10 m^3/s of fresh, running spring water. Some of this discharge was stored in the cisterns, and some was used directly from the channel, but it is apparent that more water arrived at the castle than was needed. On the east slope, between cisterns nos. 2 and 3, a channel



continued on downhill in a serpentine course to supply the winter palace of Herod on the south bank of Wadi Kelt and to irrigate the nearby plantations.

Special Structures

The storage cisterns at Cypros had to be installed as high as possible in order to minimize the toilsome portage of water up the slope into the fortress. At first sight, this level seems dictated by the topographical height of the saddle between the fortress and the adjacent

Section of run-off collecting channel, encircling hill west of Cypros.

mountain area. Provision of a bridge or pressure conduit, however, would have allowed construction of the storage cisterns higher up the slope, closer to the fort. At Cypros, the spring water traversed a long bridge, and at Machaerus a wall carried the supply channel to the

Fortress Cypros (left) and hill with collecting channel (right), from the west.





Hyrcania, channel bridge across saddle between the fortress and the Judean mountains. In foreground one of the rock cut pools.

fort. The highest saddle crossing is found at Hyrcania. Here, the natural saddle was deepened for defense purposes. Only a narrow path was left on which the foundation of the water-supply channel was laid, a substructure 112 m long and 16.80 m high. Two phases are discernible in the masonry. During the first phase, possibly of the Hasmonean period, only the four lower cisterns on the south slope could be supplied. During the second, possibly Herodian, phase twelve more cisterns higher up the slope could be filled.

Three unroofed pools measuring about 18 x 15 x 5 m deep were quarried in the rock on both sides of the path and plastered on the inside (total capacity about 4000 cubic meters). These reservoirs were filled after the cisterns on the slope of the fortress were already full. As Hyrcania received its water from the

quite frequent winter floods in the Kidron (Wadi Nur), it appears reasonable to assume that water of these reservoirs



Masonry pressurized conduit between Ras Kuneitra and Alexandrion, a very exceptional engineering feature.

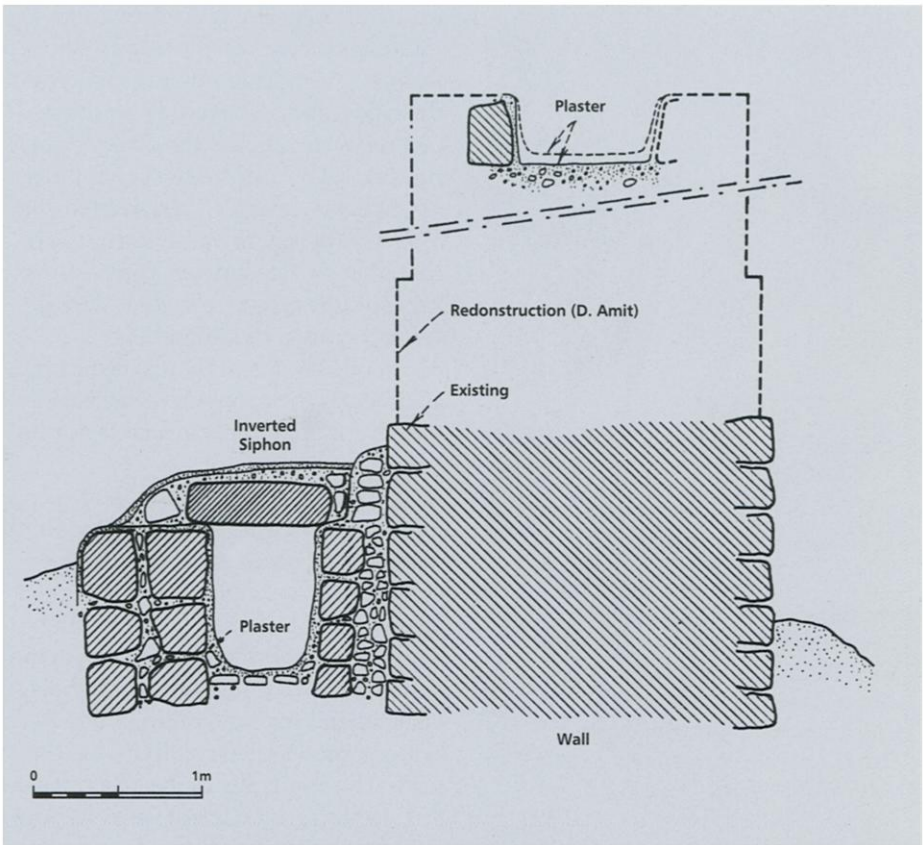
was used first and the content of the cisterns saved for later use or during a siege.

Excavation of the reservoirs served several purposes: the creation of a moat, the provision of quarried stone for the construction of the fort, and water storage.

The most interesting part of the water supply of Alexandrion is certainly the crossing of the saddle west of the fortress. Run-off from the nearby mountain Ras Kuneitra was collected by a circumferential collecting channel. The level of the lower storage cisterns could be higher because of the construction of a pressure conduit (inverted siphon) by which an aqueduct can cross a topographical depression in its course. This consisted of a masonry channel (W 0.60 m, H 0.80 m) covered by 0.30 m high stone slabs and plastered on all its interior surfaces. Perhaps because it was built on a bank of loose stone, only the beginning and the end of the inverted siphon have been preserved. The discharge end of the siphon lies 2.51 m lower than its

intake, giving a fall of 1.3% along its 190 m length, similar to the fall of the open channel upstream. As the middle part of this siphon is missing, the exact maximal pressure at its lowest point is unknown, but it was at least 5 atmospheres and perhaps reached 10 atmospheres. Assuming a water column of only 5 m, the outward pressure on a 1 m long covering slab was 30 kilonewtons (3 tons), against a counterweight of only 8 kilonewtons (0.8 tons) from the 1 m wide slab. As no clamps were found which might have fastened the cover slabs, or additional ballast, the siphon theoretically could not have functioned. After failure of the siphon, most likely a higher crossing was built.

Inverted siphons in antiquity typically were built of lead, stone, or ceramic piping. The only other slab-built inverted siphon which has been published was found at Angitia in Italy (Blake 1959: 82). At Alexandrion there is a 2.20 m wide masonry wall alongside the siphon. If this wall had been meant to carry an open channel, its masonry courses would have been horizontal. In fact, the courses are laid approximately parallel to the



Inverted siphon at Alexandrion according to Amit (1989), showing a composite cross-section of the wall and the siphon with different probable construction stages.

Desert Fortresses and their Water Systems							
Fortress	Source of Water	Collection Source	Length of Conveyance Canal	Structures	Intermediate Storage	Lift (m)	Storage in Fortress
Alexandrion	Surface runoff	Collection canal	~ 1.2 km	Inv. Syphon (1=190 m)	14 Cisterns ~4.700 m ³	~110	?
Dok	Surface flow	Diversion from Wadi	~ 500 m	-	9 Cisterns ~2.100 m ³	~40	?
Kypros	Surface run-off springs	Collection canal Conversion canal	~ 400 m ~ 14 km	Dam Bridge (135 m)	4 Cisterns ~2.000 m ³	~55	? ~4000 m ³
Hyrkania	Surface flow	Diversion from two Wadis	~ 2 km ~ 9 km	Bridge (112 m)	16 Cisterns ~16.000 m ³	~22	
Herodion	Pools of Solomon		~ 7 km	-	3 Cisterns ~2.500 m ³	~25	~ 200 m ³
Masada	Surface flow	Diversion from two Wadis	~ 350 m ~ 200 m	-	12 Cisterns ~38.500 m ³	~40 ~80	~8000 m ³
Machaerus	Surface flow	Diversion from two Wadis	?	Wall	6 Cisterns	?	?



Yehuda Peleg's field of expertise is ancient technologies. He specialises in water supply and is currently engaged in the Susita-Hippos aqueduct research project, a joint project of the Tel Aviv University and the Fachhochschule Lübeck. Peleg has explored the water systems of Caesarea, Dor and Jericho. Among his publications: *Ancient Pipelines in Israel* in *Future Currents in Aqueduct Studies* (1991), *Two Aqueducts to Dor* in *Mitteilungen des Leichtweis Instituts* (1984), and *The Water Supply of Caesarea Maritima* in *IEJ* 27 (1977).

slope of the saddle, possibly indicating that the wall was intended to carry a second inverted siphon.

Summary and Conclusion

In the mountain desert on the outskirts of the densely inhabited portion of Judaea, the Hasmonean rulers erected a number of fortresses that resemble each other both in their choice of sites facing east toward the desert and in the form of their water-supply installations. The oldest fortified vantage point was probably Dok (about 167 BCE), followed by Hyrcania (134–104 BCE). Alexander Janneus (134–107 BCE) probably founded Masada, Alexandrion, and Machaerus. Cypros, rebuilt by Herod, was most likely built by the Hasmoneans. Herodion was founded by Herod.

All the fortresses had both "consumption cisterns" inside the fort, filled by precipitation and by transport of water

from the lower storage cisterns, and "storage cisterns" on the slope of the fortress which were situated as high as the topography allowed. The storage cisterns were filled by the diversion of run-off water from wadis (Dok, Hyrcania, Masada, probably Machaerus), the collection of runoff water from an adjacent hilltop (Alexandrion, Cypros), or the capture and conveyance of spring water (Cypros, Herodion). The consumption cisterns had to be full at all times, as their contents constituted the only water-supply available during a siege.

With the exception of the "harvesting" of run-off from nearby hilltops by means of circumferential collecting channels, a seemingly local invention, all the techniques used in these water-supply systems were well known in the eastern Mediterranean area: water diversion, gravity flow conduits, conduit bridges, pressurized conduits, and cisterns. The originality of the Hasmonean engineering achievement lies in its optimal configuration of these elements to exploit in the most efficient way water resources which were distributed very unevenly in time and place. Lacking the modern scientific resources of hydrology, hydraulics, statics, and soil mechanics, and with a quite limited repertoire of building techniques, the Hasmonean and Herodian engineers nevertheless succeeded in assuring a sufficient supply of life-sustaining water to the isolated fortresses above the Jordan Valley. The prerequisites of such planning and design were a wealth of original ideas, the skilled exploitation of local conditions, good workmanship, and, not in the least, the experience accumulated during many generations of handling water in arid conditions. For their time, the installations were great technical achievements, and deserve respect even in the context of today's technology.

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Günther Garbrecht, in retirement since 1987, was Professor of Hydraulic engineering at the Middle East Technical University in Ankara and at the Technical University in Braunschweig. Complementary to his teaching and research in engineering sciences, he investigated ancient hydrotechnical systems and structures in Turkey (Tuspa, Pergamon), Egypt (Sadd el-Kafara, Fayum), Syria (Resapha), Italy (Rome, Pompeii), and Israel (Jericho). A considerable portion of his more than one hundred publications deals with the history of hydraulic engineering.

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